Blessed G/Michael Catholic Secondary School

Semester 1 Mathematics Mid1 Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** What is the limit: lim_{x -> 3} (x^2 - 9) / (x - 3)?
A. 0 **B.** 3 **C.** 6 **D.** Undefined

____ **2.** What is the 10th term of an arithmetic sequence whose first term is 4 and common difference is 3?
A. 28 **B.** 31 **C.** 34 **D.** 37

____ **3.** What is the sum of the infinite geometric series: 12 + 6 + 3 + 3/2 + ...?
A. 18 **B.** 24 **C.** 36 **D.** Diverges

____ **4.** Evaluate the limit: lim_{x -> 0} (sin(5x) / x)
A. 0 **B.** 1 **C.** 5 **D.** Undefined

____ **5.** Evaluate the limit: lim_{x -> 2} (x^2 - 4) / (x - 2)
A. 0 **B.** 2 **C.** 4 **D.** Infinity

____ **6.** What is the value of lim_{x -> infinity} (4x^3 + 2x) / (5x^3 - 7x^2 + 1)?
A. 0 **B.** 4/5 **C.** Infinity **D.** -2/7

____ **7.** A function f(x) is continuous at x = c if and only if:
A. f(c) is defined, lim_{x -> c} f(x) exists, and lim_{x -> c} f(x) = f(c)
B. f(c) exists only
C. lim_{x -> c} f(x) exists only
D. f(x) has a derivative at all points

____ **8.** What is the derivative f'(x) of the function f(x) = 3x^4 - 5x^2 + 8?
A. 12x^3 - 10x **B.** 12x^3 - 5x **C.** 7x^3 - 10x **D.** 3x^3 - 10x

____ **9.** If the common ratio of a geometric progression is r = 3 and the first term is a1 = 2, what is the 5th term?
A. 54 **B.** 162 **C.** 486 **D.** 243

____ **10.** Evaluate lim_{x -> 0} [(e^x - 1) / x]:
A. 0 **B.** 1 **C.** e **D.** Undefined

____ **11.** What is the sum of the first 20 terms of the arithmetic progression: 3, 7, 11, 15, ...?
A. 820 **B.** 800 **C.** 840 **D.** 780

____ **12.** Find the limit: lim_{x -> 0} [(sqrt(1 + x) - 1) / x]
A. 1/2 **B.** 1 **C.** 0 **D.** 2

____ **13.** Evaluate lim_{x -> 1} (x^3 - 1) / (x - 1):
A. 3 **B.** 1 **C.** 0 **D.** Infinity

____ **14.** What is the sum to infinity of the geometric series sum_{n=1}^infinity (3 / 4^n)?
A. 1 **B.** 3/4 **C.** 4/3 **D.** 1/3

____ **15.** If lim_{x -> 2} [f(x) - 5] / (x - 2) = 3, what is the value of lim_{x -> 2} f(x)?
A. 5 **B.** 3 **C.** 0 **D.** 8

____ **16.** Evaluate lim_{x -> 0} [tan(3x) / sin(2x)]:
A. 3/2 **B.** 2/3 **C.** 1 **D.** 6

____ **17.** Which of the following sequences is divergent?
A. an = n / (n + 1) **B.** an = 1 / 2^n
C. an = (-1)^n * n **D.** an = 3 + 1/n

____ **18.** If a sequence has general term an = (2n - 1) / (3n + 5), what is lim_{n -> infinity} an?
A. 2/3 **B.** -1/5 **C.** 0 **D.** Infinity

____ **19.** The product of three consecutive terms in a geometric progression is 216. What is the middle term?
A. 6 **B.** 3 **C.** 12 **D.** 18

____ **20.** Find the value of k that makes f(x) continuous at x = 3: f(x) = (x^2 - 9)/(x - 3) for x != 3, and f(3) = k.
A. 6 **B.** 3 **C.** 0 **D.** 9

____ **21.** Evaluate lim_{x -> 0} (1 - cos(x)) / x^2:
A. 1/2 **B.** 0 **C.** 1 **D.** 2

____ **22.** What is the sum of the series sum_{n=1}^10 (2n + 1)?
A. 120 **B.** 110 **C.** 100 **D.** 130

____ **23.** Evaluate lim_{x -> infinity} [sqrt(x^2 + 4x) - x]:
A. 2 **B.** 0 **C.** 4 **D.** Infinity

____ **24.** Which of the following describes an arithmetic-geometric sequence?
A. A sequence where terms are the product of corresponding terms of an AP and a GP
B. A sequence where differences are constant only
C. A sequence with alternating signs only
D. A Fibonacci sequence

____ **25.** What is the limit of the sequence an = (1 + 1/n)^n as n -> infinity?
A. e (~2.718) **B.** 1 **C.** Infinity **D.** 0